Enhao Zhang

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Research Interests

- AI-Driven Data Management Systems, Databases, Video Analytics, Large Language Models, Agentic AI

Education

- **University of Washington** Seattle, WA
  Ph.D. in Computer Science *Sept. 2020 – Present*
  - Advisors: Prof. **Magdalena Balazinska** and Prof. **Ranjay Krishna**
- **University of Michigan** Ann Arbor, MI
  Bachelor of Science Engineering in Computer Science *Sept. 2018 – Apr. 2020*
  - Overall GPA: 4.00/4.00
  - Advisors: Prof. **Nikola Banovic** and Prof. **Michael Cafarella**
- **Shanghai Jiao Tong University** Shanghai, China
  Bachelor of Science in Electrical and Computer Engineering *Sept. 2015 – Aug. 2020*
  - Overall GPA: 3.97/4.00 (Ranking: 1^{st}/202)

Publications

- KathDB: Explainable Multimodal Database Management System with Human-AI Collaboration. Guorui Xiao, **Enhao Zhang**, Nicole Sullivan, Will Hansen, Magdalena Balazinska. CIDR 2026.
- Optimizing Sequential Multi-Step Tasks with Parallel LLM Agents. **Enhao Zhang**, Erkang Zhu, Gagan Bansal, Adam Fourney, Hussein Mozannar, Jack Gerrits. MAS@ICML 2025. [Paper]
- Self-Enhancing Video Data Management System for Compositional Events with Large Language Model. **Enhao Zhang**, Nicole Sullivan, Brandon Haynes, Ranjay Krishna, Magdalena Balazinska. SIGMOD 2025. [Paper][Code]
- Bootstrapping Compositional Video Query Synthesis with Natural Language and Previous Queries from Users. Manasi Ganti, **Enhao Zhang**, Magdalena Balazinska. HILDA@SIGMOD 2025. **Paper selected as "Best of HILDA" and invited to the Special Issue on Human-in-the-Loop Data Analytics (HILDA) in the *Information Systems Journal*.** [Paper]
- VOCALExplore: Pay-as-You-Go Video Data Exploration and Model Building. Maureen Daum, **Enhao Zhang**, Dong He, Stephen Mussmann, Brandon Haynes, Ranjay Krishna, Magdalena Balazinska. VLDB 2024. [Paper][Code]
- EQUI-VOCAL: Synthesizing Queries for Compositional Video Events from Limited User Interactions. **Enhao Zhang**, Maureen Daum, Dong He, Brandon Haynes, Ranjay Krishna, Magdalena Balazinska. VLDB 2023. [Paper][Code]
- EQUI-VOCAL Demonstration: Synthesizing Video Queries from User Interactions. **Enhao Zhang**, Maureen Daum, Dong He, Manasi Ganti, Brandon Haynes, Ranjay Krishna, Magdalena Balazinska. VLDB 2023 Demo. [Paper]
- VOCAL: Video Organization and Interactive Compositional AnaLytics. Maureen Daum*, **Enhao Zhang***, Dong He, Magdalena Balazinska, Brandon Haynes, Ranjay Krishna, Apryle Craig, Aaron Wirsing. CIDR 2022. (* indicates equal contributions) [Paper]
- Method for Exploring Generative Adversarial Networks (GANs) via Automatically Generated Image Galleries. **Enhao Zhang**, Nikola Banovic. CHI 2021. [Paper][Website]

Honors and Awards

- **Madrona Prize** (Recognizing the most commercializable research project) (☞ Link), Paul G. Allen School of Computer Science & Engineering, UW, 2022
- **Cheng Family Scholarship**, Joint Institute, Shanghai Jiao Tong University, 2018
- **Interdisciplinary Contest in Modeling**, Honorable Mention, 2017
- **Distinguished Academic Achievement Award** (Academic performance in the top 2% of class), Joint Institute, Shanghai Jiao Tong University, 2016
- **Undergraduate National Scholarship** (Top 7 students in Joint Institute), Ministry of Education of People's Republic of China, 2016

Research and Work Experience

- **University of Washington** Seattle, WA
  Research assistant | Advisors: Prof. Magdalena Balazinska, Prof. Ranjay Krishna *Sept. 2020 – Present*
  - **VOCAL-UDF**: Designed and implemented a video query processing system that supports compositional queries by dynamically generating missing operators as user-defined functions. Developed a unified execution model integrating Python programs and distilled ML models as query operators, enabling extensible query execution over heterogeneous compute components. Built infrastructure for automated operator generation and validation, using active learning to select optimal implementations and improve reliability of LLM-generated code.
  - **EQUI-VOCAL**: Built a query synthesis and execution system for complex video analytics workloads. Proposed a spatio-temporal scene graph data model and query language to express compositional events over large video collections. Developed efficient search and pruning strategies (beam search + active learning) to synthesize executable queries while reducing computational cost.
  - **VOCALExplore**: Developed an interactive system that supports users in building domain-specific models over videos, with a dynamic sampling strategy to maximize model quality, a rising bandit optimization algorithm to improve training efficiency, and a task scheduler to ensure low user-visible latency.
- **Microsoft Research** Redmond, WA
  Research intern, AI Frontiers Team | Mentor: Erkang (Eric) Zhu *Jun. 2024 - Sept. 2024*
  - Investigated performance optimization techniques for LLM-driven multi-agent systems executing multi-step workflows. Designed parallel execution strategies with early termination and aggregation, reducing end-to-end latency and improving task completion rates.
- **Snowflake** San Mateo, CA
  PhD software engineer intern, SQL Query Language Team | Mentor: Dmitry Lychagin *Jun. 2023 - Sept. 2023*
  - Designed and implemented NULL semantics improvements for SQL functions and operators across the Snowflake query engine. Contributed to multiple stages of the query processing pipeline, including parsing, semantic validation, query optimization, and execution. Implemented production code in Java and C++ within Snowflake's distributed query engine.
- **University of Michigan** Ann Arbor, MI
  Research intern | Advisor: Prof. Nikola Banovic *Sept. 2019 – Sept. 2020*
  - **GAN Explorer:** Designed an interactive tool for Generative Adversarial Network (GAN) exploration and evaluation, where users can assess capabilities and limitations of a GAN via interactive visual examination. Used a Markov Chain Monte Carlo (MCMC) method for automated image gallery generation, which enabled quick creation of many diverse, photo-realistic image galleries to support qualitative evaluation of GANs.

Mentoring Experience

- **Undergraduate students:** Will Hansen, Manasi Ganti, Brian Yao, Chongjiu Gao, Lyons (Daoyi) Lu, Yichi Zhang
- **High school students:** Anish Chaudhuri, Parie Kumar

Professional Service

- **Reviewer** – CHI 2022, CSCW 2022, MAS@ICML 2025, CHI 2026
- **Session host** – CS Education Week, University of Washington

Tutoring Experience

- **TA, CSE 444: Database Systems Internals**, University of Washington *Winter 2023, 2025*
- **TA, VY200: Academic Writing II**, Shanghai Jiao Tong University *Spring 2017*
- **TA, VY100: Academic Writing I**, Shanghai Jiao Tong University *Fall 2016*

Skills

- **Programming Languages:** Python, C/C++, Java, SQL, JavaScript, HTML
- **Frameworks & Libraries:** PyTorch, NVIDIA DALI, scikit-learn, pandas, Django, PyTorch Lightning, PEFT
- **Other Tools:** PostgreSQL, DuckDB, AWS, Docker, Slurm, Hugging Face, Git, LaTeX